

ATHARVA MAHAJAN

Generative Models & Reinforcement Learning for Physical Systems · M.S. University of Michigan

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Profile

Machine-learning researcher building **generative models and closed-loop control policies for high-dimensional physical systems**. First author of *Policy-DRIFT* (conditional flow matching, reward-guided sampling, deep RL), which reaches **48.95% drag reduction at 37× lower actuation energy**; prior work spans neural surrogates, DNS/LES turbulence, and **450× CUDA acceleration** of a CFD solver.

Education

University of Michigan - *M.S. Mechanical Engineering, GPA 3.98/4; focus: generative modeling* Aug 2025 – May 2027
BITS Pilani, Hyderabad - *B.E. Mechanical Engineering (CGPA 9.16/10) · M.Sc. Physics* Nov 2020 – May 2025

Technical Skills

Machine Learning: Generative modeling (conditional flow matching, diffusion, guided sampling), deep reinforcement learning, reward modeling, CNNs/RNNs/autoencoders, neural surrogates & reduced-order modeling, scientific & physics-informed ML, closed-loop control
Languages & Frameworks: Python, C/C++, CUDA, Fortran, MATLAB, Bash · PyTorch, TensorFlow, NVIDIA Nsight, GPU/HPC clusters, Git, $\LaTeX$
Simulation & Hardware: DNS/LES, spectral & meshfree solvers, boundary element methods, Dedalus, ANSYS Fluent, COMSOL, SolidWorks, Fusion 360, Arduino / embedded control

Research Experience: Machine Learning & Simulation

Policy-DRIFT: Reward-Guided Generative Control of Turbulence Aug 2025 – Present
First author · Graduate Researcher, Prof. Ricardo Vinuesa, University of Michigan · [arXiv:2605.14022](https://arxiv.org/abs/2605.14022) Ann Arbor, MI

- Trained a **conditional flow-matching** generative model over a multi-regime flow manifold and steered its sampling trajectories with **Terminal Reward Guidance** to synthesize physically realizable, reward-maximizing targets.
- Trained a lightweight **deep-RL policy** to track those targets, reaching **48.95% drag reduction** at $Re_\tau = 180$ (**+16.2% over the DRL benchmark**) at **37× lower actuation energy**.
- Built the end-to-end **PyTorch/GPU** stack over 22,000+ DNS snapshot pairs; the deployed controller runs from wall-only sensing with **no generative queries at inference**.

Deep Learning for Fluid Mechanics - Research & Teaching Assistant Jan 2025 – Present
University of Michigan · Flowthermolab (industry collaboration) · Siemens (Stockholm) Ann Arbor, MI

- Designed and taught a hands-on deep-learning curriculum (MLPs, RNNs, autoencoders, CNNs) for 30+ graduate students and practicing engineers.
- Authored the course's core **PyTorch/Jupyter** notebook suite on **flow-field reconstruction, reduced-order modeling, and data-driven turbulence closure**; delivered the applied version to Siemens engineers in Stockholm.

Upstream Pressure-History Effects on Adverse-Pressure-Gradient TBLs Jun 2024 – Jun 2025
Advisor: Prof. Ricardo Vinuesa, KTH Royal Institute of Technology Stockholm, Sweden

- Ran **DNS/LES** campaigns on turbulent boundary layers under varying upstream pressure histories, quantifying non-local memory effects via spectral densities, Reynolds-stress budgets, and β -evolution. Published in *Int. J. Heat and Fluid Flow* (2025).

GPU-Accelerated Meshfree CFD Solver (CUDA) Nov 2022 – Dec 2023
Advisor: Dr. Anil Nemili, BITS Pilani Hyderabad, India

- Built a 3D meshfree Euler solver in Fortran/CUDA and drove a **450× GPU speedup** via kernel restructuring, memory-coalescing, and NVIDIA Nsight-guided profiling.

Autophoretic Particle Dynamics, BITS Pilani - BEM/COMSOL microswimmer modeling May 2023 – May 2024

- Modeled self-propelled microparticle locomotion near deformable interfaces using boundary element methods and COMSOL, with applications to targeted drug delivery and environmental remediation.

Underwater Pitching-Airfoil Propulsion, Experimental - Dr. Pardha Saradhi, BITS Pilani Feb 2021 – Dec 2023

- Built a pitching-airfoil test rig for bio-inspired thrust generation with MATLAB force/torque acquisition and dye visualization for parametric efficiency mapping.

Professional Experience

Applied Aero Labs Pvt Ltd - Co-founder & Technical Lead Apr 2023 – May 2025

- Co-founded an aerodynamic R&D venture building India's first portable, multi-velocity fan-array wind tunnel; secured **\$12,000 in government funding** and owned mechanical design, embedded control/electronics integration, and client validation.

Leadership & Teaching

President, SEDS BPHC (2023): led the technical roadmap of a 120-member student space org and two satellite projects.

Structures Lead, Hyperloop India: pod-frame and vacuum-seal design. **TA, Fluid Mechanics:** 200 undergraduates; tutorials, CFD labs, mentoring.